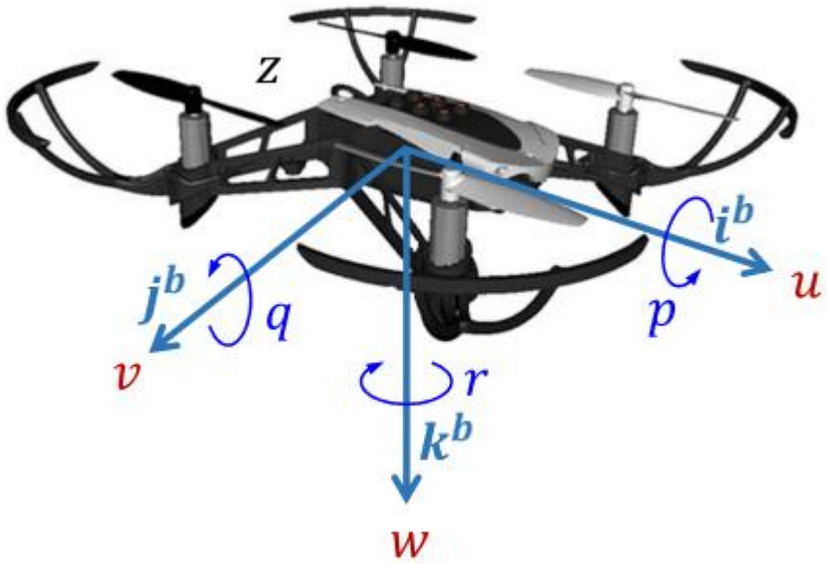
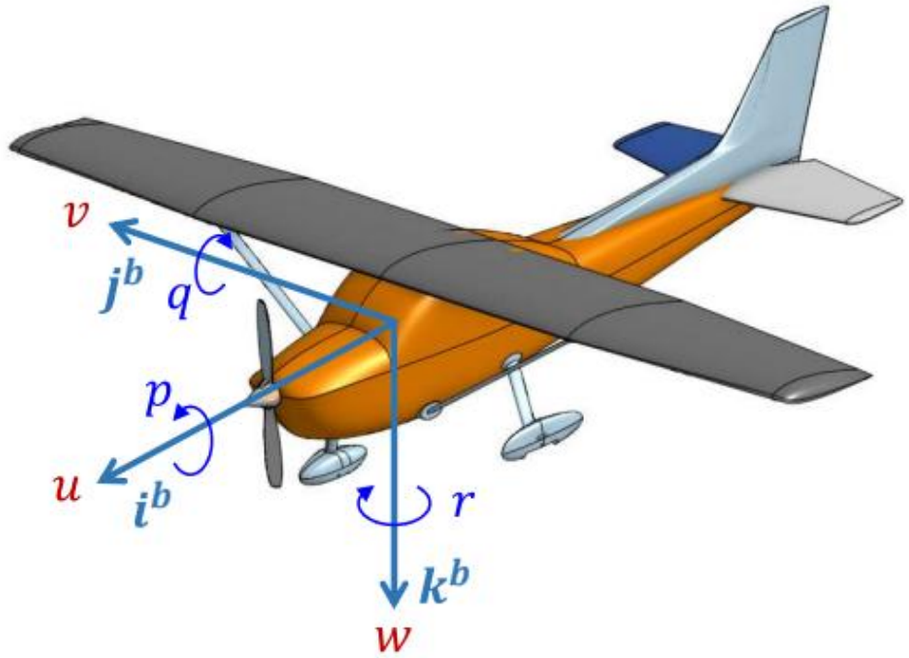
Topic 3: Kinematics and Dynamics

How do forces and moments change flight?

Translational motion and rotational motion





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Learning goals for this session

Explain why rotating body axes introduce extra terms.

Use Newton–Euler equations to calculate accelerations.

Connect body velocities to position and attitude.

Identify what changes between multirotor and fixed-wing models.

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Kinematics and Dynamics

Kinematics describes motion

Position, orientation and their rates of change.

Dynamics explains changes in motion

Forces change linear momentum. Moments change angular momentum.

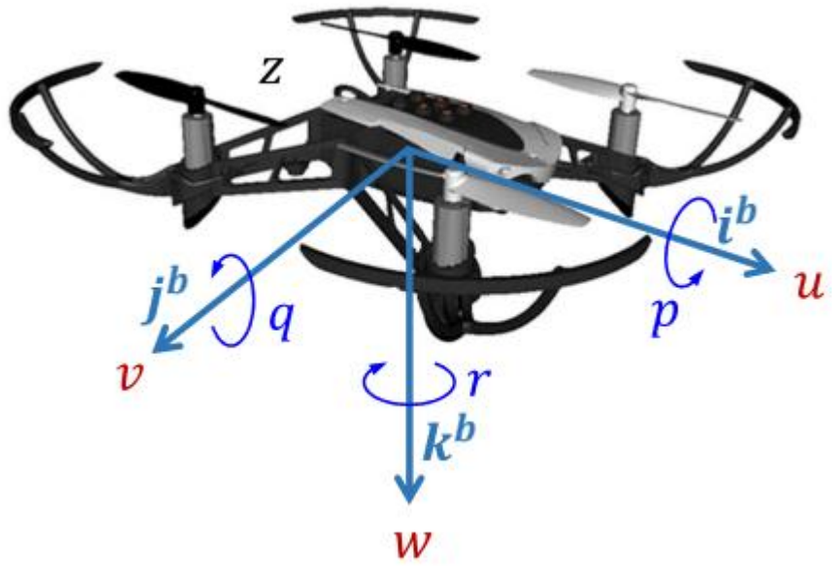
Discussion: can a UAV have zero acceleration while moving?

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Frames and symbols

Body b: forward, right, down

NED n: north, east, down



u, v, w: body-axis velocity components (m/s)

p, q, r: body-axis angular rates (rad/s)

ϕ, θ, ψ: roll, pitch and yaw angles

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Model assumptions

Rigid aircraft with constant mass and inertia.

Body origin at the centre of mass.

Local NED frame treated as inertial.

All vectors in a cross product use the same coordinate frame.

F^b includes all external forces. M^b is the external moment about the centre of mass.

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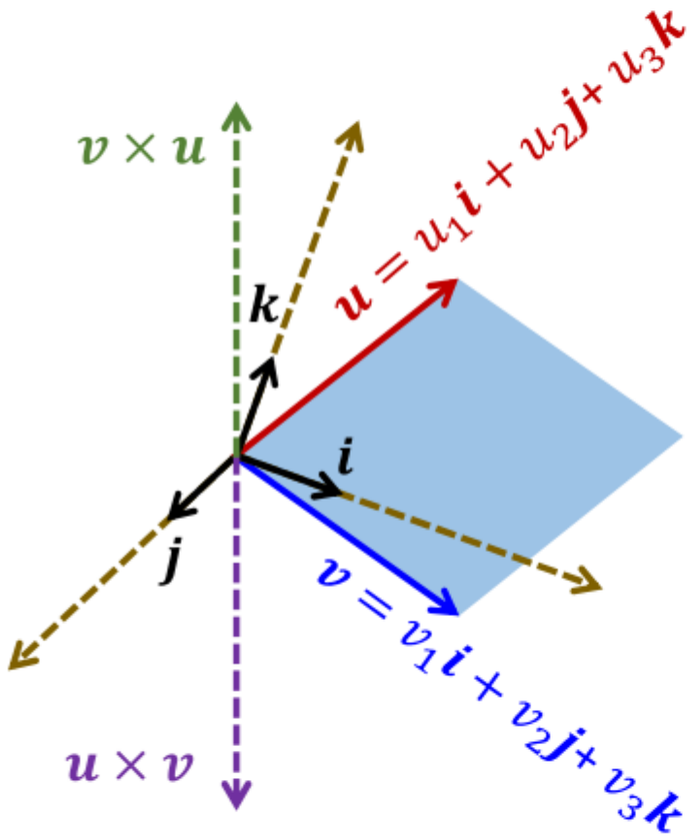
Cross product: Order matters

$$a \times b = \begin{bmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{bmatrix}$$

$$a \times b = -(b \times a)$$

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Cross product



$$u \times v = \begin{vmatrix} i & j & k \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

$$= \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} i - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} j + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} k$$

$$u \times v = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}$$

$$v \times u = \begin{bmatrix} v_2 u_3 - v_3 u_2 \\ v_3 u_1 - v_1 u_3 \\ v_1 u_2 - v_2 u_1 \end{bmatrix}$$

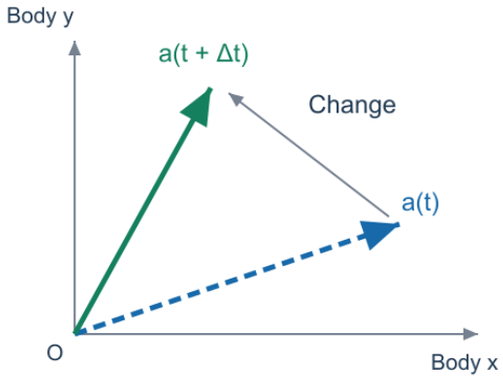
$$\|u \times v\| = \|v \times u\| = \|u\| \, \|v\| sin(\theta)$$

Cross product, vector product, or directed area product. The cross product of two linearly independent vectors is a vector perpendicular to both with a magnitude equivalent to the area made between them.

A vector can change as its axes rotate

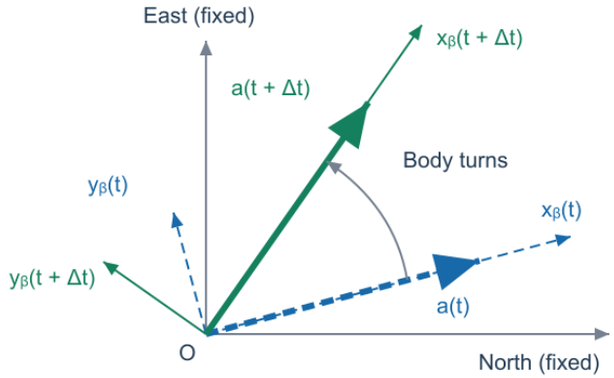
Two contributions to a vector's change

1. Vector changes within the body frame



Axes held fixed; vector changes.

2. Body frame rotates



Vector stays attached along body x.
Its direction still changes in NED.

Ground-observed change = component change + rotation of the axes

$(da/dt)^b$ inertial = $\dot{a}_\beta + \omega_\beta \times a_\beta$

Dashed: earlier. Solid: later. Same origin. Equation terms all expressed in body coordinates.

Two contributions to the derivative:

1. Components change relative to the body.
2. The body axes rotate relative to NED.

A vector fixed to the aircraft can still change direction in NED.

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Derivative in a rotating frame

$$\left(\frac{da}{dt}\right)_n^b = \dot{a}_b + \omega_b \times a_b$$

a_b: components of the vector in body axes

$\omega_b = [p, q, r]^T$: body angular velocity

If the body does not rotate, $\omega_b = 0$ and the extra term vanishes.

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Worked example: rotating direction

A unit direction is fixed along body x. The body turns about z.

$$a_b = \begin{bmatrix}1\\0\\0\end{bmatrix}, \omega_b = \begin{bmatrix}0\\0\\2\end{bmatrix}, \dot{a}_b = 0$$

$$\left(\frac{da}{dt}\right)_n^b = \begin{bmatrix}0\\0\\2\end{bmatrix} \times \begin{bmatrix}1\\0\\0\end{bmatrix} = \begin{bmatrix}0\\2\\0\end{bmatrix}$$

The direction changes towards positive body y. Angular rate: 2 rad/s.

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Forces determine acceleration

First apply Newton's law, then express it in body coordinates.

Which frame does Newton's law use?

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Newton's second law

$$F = m(\frac{dV}{dt})_n$$

$$F^b = m(\dot{v}_b + \omega_b \times v_b)$$

$v_b = [u, v, w]^T$

The force vector and velocity components now use the same body axes.

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Body-axis acceleration equation

$$\frac{F^b}{m} = \dot{v}_b + \omega_b \times v_b$$

$$\dot{v}_b = \frac{F^b}{m} - \omega_b \times v_b$$

The cross-product term accounts for the rotating coordinates.

Check: when p = q = r = 0, the result becomes F^b/m.

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Three translational equations

$$\dot{u} = rv - qw + \frac{F_x}{m}$$

$$\dot{v} = pw - ru + \frac{F_y}{m}$$

$$\dot{w} = qu - pv + \frac{F_z}{m}$$

F_x, F_y, F_z are total force components in body axes (N).

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Gravity and thrust

Level UAV with no body rotation. Ignore aerodynamic drag.

$$F_z = mg - T$$

$$\dot{w} = g - \frac{T}{m}$$

Positive w and positive vertical acceleration point downwards.

If T > mg, which sign should the acceleration have?

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Worked example: vertical acceleration

A level 2.00 kg UAV produces 24.0 N upward thrust. Ignore drag and body rotation. Use g = 9.81 m/s².

$$F_z = 2.00(9.81) - 24.0 = -4.38$$

Net force: −4.38 N in body z

$$\dot{w} = \frac{-4.38}{2.00} = -2.19$$

Acceleration: 2.19 m/s² upwards.

Does this tell us whether the UAV is already moving upwards?

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Your turn: body-axis acceleration

At one instant, m = 2.00 kg and all vectors below use body axes.

$$v_b = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}, \omega_b = \begin{bmatrix} 0 \\ 0 \\ 0.5 \end{bmatrix}, F^b = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

Velocity in m/s, angular rate in rad/s, total force in N.

Calculate $\dot{u}$, $\dot{v}$ and $\dot{w}$. Explain why $\dot{v}$ can be nonzero when $F_y = 0$.

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Solution: body-axis acceleration

$$\omega_b \times v_b = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$$

$$\dot{v}_b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}$$

$\dot{u}$ = 1.00, $\dot{v}$ = −2.00, $\dot{w}$ = 0 m/s².

The y component changes because the coordinate axes rotate.

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2. Rotational dynamics

Force F changes linear momentum.

Moment M changes angular momentum h.

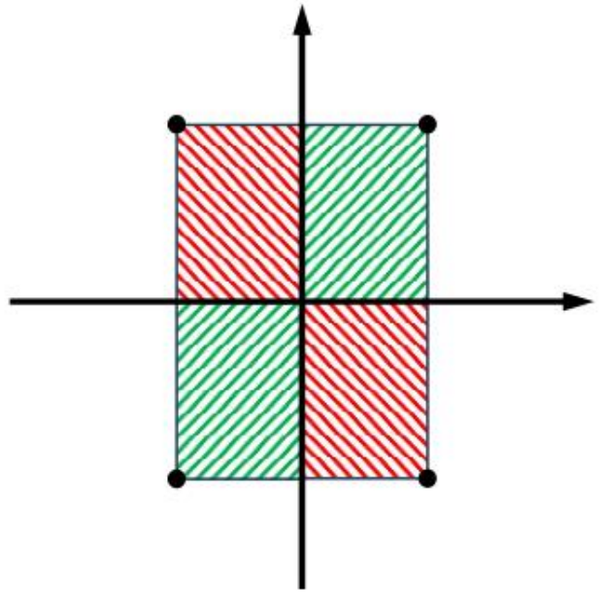
$$h_b = J\omega_b$$

J: inertia tensor (kg m^2). M: external moment (N m).

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Inertia and mass distribution

Inertia describes resistance to angular acceleration.



For body axes aligned with principal axes:

$$J = \begin{bmatrix} J_x & 0 & 0 \\ 0 & J_y & 0 \\ 0 & 0 & J_z \end{bmatrix}$$

Discussion: how does moving a battery farther from an axis affect its inertia?

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Euler's rigid-body equation

$$M^b = J\dot{\omega}_b + \omega_b \times (J\omega_b)$$

$$\dot{\omega}_b = J^{-1}(M^b - \omega_b \times (J\omega_b))$$

Applied moment and existing rotation both affect the body rates.

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Principal-axis rotational equations

$$\dot{p} = \frac{M_x + (J_y - J_z)qr}{J_x}$$

$$\dot{q} = \frac{M_y + (J_z - J_x)pr}{J_y}$$

$$\dot{r} = \frac{M_z + (J_x - J_y)pq}{J_z}$$

Body rates p, q, r are in rad/s. Their derivatives are in rad/s².

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Worked example: roll acceleration

$J_x = 0.020$ kg m², $M_x = 0.060$ N m and $q = r = 0$.

$$\dot{p} = \frac{M_x}{J_x} = \frac{0.060}{0.020} = 3.00$$

Roll angular acceleration: 3.00 rad/s².

What happens if J_x doubles while M_x stays the same?

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Your turn: rotational coupling

Principal moments: $J_x = 0.020$, $J_y = 0.030$, $J_z = 0.040$ kg m^2.

Body rates: p = 0, q = 1.00, r = 2.00 rad/s.

Moments: $M_x = 0.060$ N m, $M_y = M_z = 0$.

Calculate $\dot{p}$, $\dot{q}$ and $\dot{r}$ at this instant.

Compare $\dot{p}$ with the previous single-axis example.

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Solution: rotational coupling

$$(J_y - J_z)qr = (0.030 - 0.040)(1)(2) = -0.020$$

Coupling contribution: −0.020 N m

$$\dot{p} = \frac{0.060 - 0.020}{0.020} = 2.00$$

$\dot{p}$ = 2.00 rad/s². Here $\dot{q}$ = $\dot{r}$ = 0.

The existing pitch and yaw rates change the roll response.

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3. Complete aircraft model

Position: p_N, p_E, p_D

Velocity: u, v, w

Attitude: ϕ, θ, ψ

Body rates: p, q, r

$$\dot{x} = f(x, U)$$

Here U contains the applied force and moment components.

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Position rates in NED

$$\begin{bmatrix} \dot{p}_N \\ \dot{p}_E \\ \dot{p}_D \end{bmatrix} = R_b^n \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

The rotation matrix maps body velocity components into NED.

Quick check: level, facing east, moving forward at 5 m/s.

What are the north, east and down velocity components?

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Body rates and Euler-angle rates

p, q, r describe rotation about body axes.

$\dot{\phi}$, $\dot{\theta}$, $\dot{\psi}$ describe changes in Euler angles.

$$\phi = \theta = 0: \qquad \dot{\phi} = p, \dot{\theta} = q, \dot{\psi} = r$$

This simple equality does not hold at a general attitude.

The ZYX Euler-rate mapping becomes singular at $\theta = \pm 90°$.

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Quaternion attitude: what changes?

A unit quaternion η stores orientation using four components.

$$\eta = [\eta_0, \eta_1, \eta_2, \eta_3]^T, \|\eta\| = 1$$

It avoids the Euler-angle singularity in attitude propagation.

The translational and rotational dynamics stay the same.

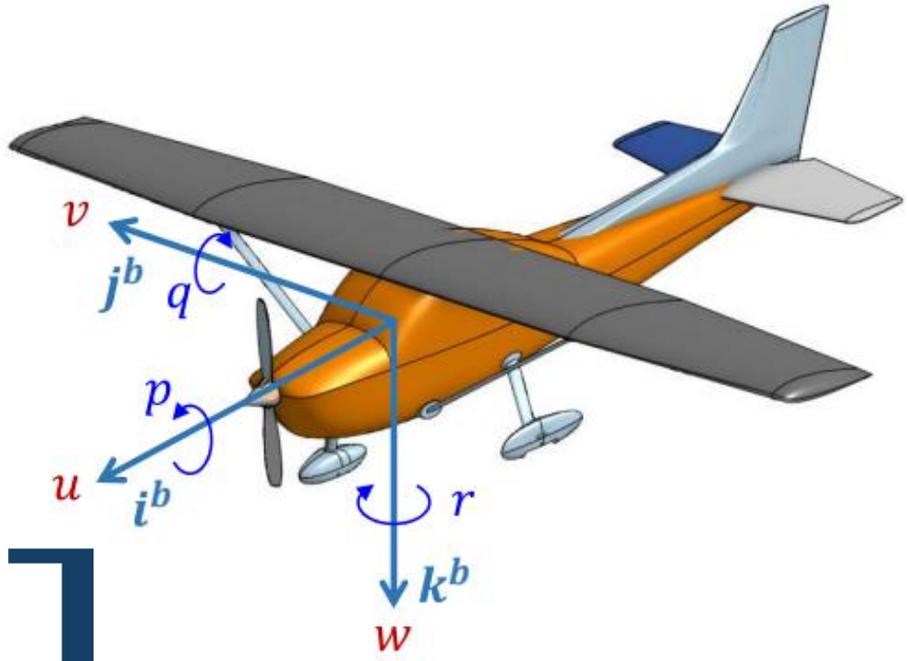
Convention here: scalar first, Hamilton product, body-to-NED rotation.

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Fixed-wing and multirotor models

Both obey the same rigid-body laws.

The differences lie in inertia and how forces and moments arise.



$$J = \begin{bmatrix} J_x & 0 & -J_{xz} \\ 0 & J_y & 0 \\ -J_{xz} & 0 & J_z \end{bmatrix}$$

A symmetric fixed-wing model may retain J_{xz}.

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Pair task: Assemble a vertical model

A level UAV moves vertically. Use z positive downwards.

Assume constant mass m, no rotation and no drag.

Choose states z and w. Input T is total upward thrust.

1. Write the two state equations.

2. State the conditions for hover.

3. Does T = mg always mean the UAV is stationary?

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Solution: A two-state model

$$\dot{z} = w$$

$$\dot{w} = g - \frac{T}{m}$$

Hover: w = 0 and T = mg.

If T = mg but w ≠ 0, vertical velocity stays constant in this model.

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Summary

A rotating frame adds a cross-product term to vector derivatives.

Total forces determine translation. Moments and inertia determine rotation.

Kinematics connects body rates and velocities to attitude and position.

Both aircraft types use the same mechanics, with different force and inertia models.

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Exit check

1. Does a body-fixed vector always have zero inertial derivative?

2. With z downwards, what does negative vertical acceleration mean?

3. Why can M_x/J_x give the wrong roll acceleration?

4. Can the same rigid-body laws describe fixed-wing aircraft?

Explain your reasoning to a partner.

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Appendix: supporting equations

Transport theorem and component expansion

ZYX position and attitude mappings

Quaternion propagation and rotation matrix

Fixed-wing inertia coupling

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Transport theorem: component derivation

$$a = a_1 i_b + a_2 j_b + a_3 k_b$$

$$\left(\frac{di_b}{dt}\right)_n = \omega \times i_b$$

$$\left(\frac{da}{dt}\right)_n = \dot{a}_1 i_b + \dot{a}_2 j_b + \dot{a}_3 k_b + \omega \times a$$

The first three terms describe component changes. The final term describes rotation of the basis.

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Body-to-NED rotation matrix

ZYX yaw–pitch–roll. Use cα = cos α and sα = sin α.

$$R_b^n = \begin{bmatrix} c_\theta c_\psi & s_\varphi s_\theta c_\psi - c_\varphi s_\psi & c_\varphi s_\theta c_\psi + s_\varphi s_\psi \\ c_\theta s_\psi & s_\varphi s_\theta s_\psi + c_\varphi c_\psi & c_\varphi s_\theta s_\psi - s_\varphi c_\psi \\ -s_\theta & s_\varphi c_\theta & c_\varphi c_\theta \end{bmatrix}$$

Check $R^T R = I$ and $\det(R) = +1$.

The reverse transformation uses the transpose of this matrix.

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ZYX Euler-angle rates

$$\begin{bmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & sin\varphi tan\theta & cos\varphi tan\theta \\ 0 & cos\varphi & -sin\varphi \\ 0 & sin\varphi/cos\theta & cos\varphi/cos\theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

Use radians in differential equations.

This mapping requires cos θ ≠ 0.

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Quaternion propagation

η maps body to NED. ω = [0, p, q, r]^T contains body rates.

$$\dot{\eta} = \frac{1}{2}\eta \otimes \omega$$

$$\dot{\eta} = \frac{1}{2}\begin{bmatrix} 0 & -p & -q & -r \\ p & 0 & r & -q \\ q & -r & 0 & p \\ r & q & -p & 0 \end{bmatrix}\eta$$

Maintain ||η|| = 1 during numerical integration.

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Quaternion rotation matrix

For a unit quaternion $\eta = [\eta_0, \eta_1, \eta_2, \eta_3]^T$:

$$R_b^n = \begin{bmatrix} 1-2(\eta_2^2+\eta_3^2) & 2(\eta_1\eta_2-\eta_0\eta_3) & 2(\eta_1\eta_3+\eta_0\eta_2) \\ 2(\eta_1\eta_2+\eta_0\eta_3) & 1-2(\eta_1^2+\eta_3^2) & 2(\eta_2\eta_3-\eta_0\eta_1) \\ 2(\eta_1\eta_3-\eta_0\eta_2) & 2(\eta_2\eta_3+\eta_0\eta_1) & 1-2(\eta_1^2+\eta_2^2) \end{bmatrix}$$

Position rates still follow from rotating body velocity into NED.

The quaternion represents orientation, not position.

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Fixed-wing inertia coupling

$$J = \begin{bmatrix} J_x & 0 & -J_{xz} \\ 0 & J_y & 0 \\ -J_{xz} & 0 & J_z \end{bmatrix}$$

$$J^{-1} = \begin{bmatrix} J_z/\Gamma & 0 & J_{xz}/\Gamma \\ 0 & 1/J_y & 0 \\ J_{xz}/\Gamma & 0 & J_x/\Gamma \end{bmatrix}$$

$$\Gamma = J_x J_z - J_{xz}^2 > 0$$

Use $\dot{\omega}_b = J^{-1}[M^b - \omega_b \times (J\omega_b)]$.

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Fixed-wing component equations

Define the three components of $M^b - \omega_b \times (J\omega_b)$:

$$A = M_x + (J_y - J_z)qr + J_{xz}pq$$

$$B = M_y + (J_z - J_x)pr - J_{xz}(p^2 - r^2)$$

$$C = M_z + (J_x - J_y)pq - J_{xz}qr$$

$$\dot{p} = \frac{J_z A + J_{xz} C}{\Gamma}, \dot{q} = \frac{B}{J_y}, \dot{r} = \frac{J_{xz} A + J_x C}{\Gamma}$$

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